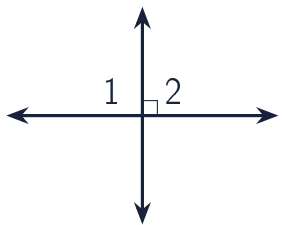
Lesson 3.4

Perpendicular Lines

Right angles, and the shortest distance to a line

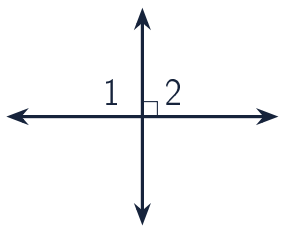
fangmath.com · YouTube @fangmathcom

Perpendicular lines



Perpendicular lines meet at a **right angle**: $a \perp b$

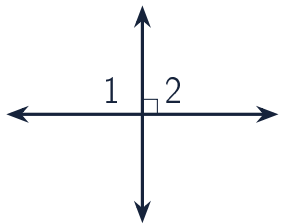
Perpendicular lines



Perpendicular lines meet at a **right angle**: $a \perp b$

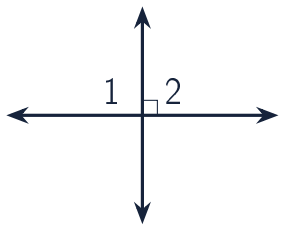
The little **square** in a picture marks a right angle.

Linear pair of congruent angles



$\angle 1$ and $\angle 2$ form a linear pair:
$m\angle 1 + m\angle 2 = 180°$

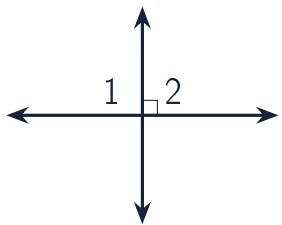
Linear pair of congruent angles



$\angle 1$ and $\angle 2$ form a linear pair:
$m\angle 1 + m\angle 2 = 180°$

If $\angle 1 \cong \angle 2$: each is $180 \div 2 = 90°$

Linear pair of congruent angles



$\angle 1$ and $\angle 2$ form a linear pair:
$m\angle 1 + m\angle 2 = 180°$

If $\angle 1 \cong \angle 2$: each is $180 \div 2 = 90°$

Two lines that form a linear pair of congruent angles are **perpendicular**.

Two lines perpendicular to the same line

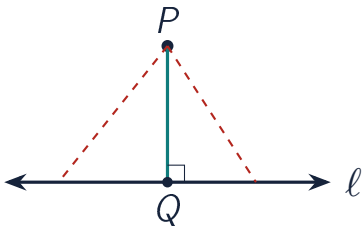
In a plane: $a \perp t$ and $b \perp t \Rightarrow a \parallel b$

Two lines perpendicular to the same line

In a plane: $a \perp t$ and $b \perp t \Rightarrow a \parallel b$

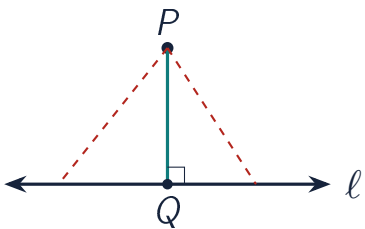
Why: corresponding angles are both $90°$, so they are congruent.

Distance from a point to a line



Drop a segment from P straight down to ℓ: $\overline{PQ} \perp \ell$

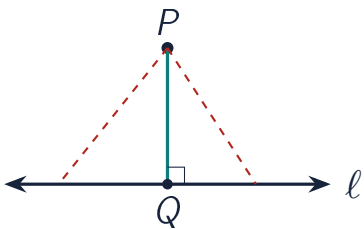
Distance from a point to a line



Drop a segment from P straight down to ℓ: $\overline{PQ} \perp \ell$

Every slanted path is **longer** than PQ.

Distance from a point to a line



Drop a segment from P straight down to ℓ: $\overline{PQ} \perp \ell$

Every slanted path is **longer** than PQ.

Distance from a point to a line = length of the **perpendicular** segment.

Watch out

Watch out! Only the **perpendicular** segment measures the distance to a line — not a slanted one.

Your turn! Pause the video and try it.

Point $P(3,5)$ and the line $y = 1$. What is the distance from P to the line?

Your turn! Pause the video and try it.

Point $P(3,5)$ and the line $y = 1$. What is the distance from P to the line?

Go straight down to $(3,1)$: distance $= 5 - 1 = 4$

Remember

- **Perpendicular** lines meet at 90°.
- Linear pair of congruent angles $\Rightarrow$ perpendicular.
- Distance to a line = the **perpendicular** segment, the shortest path.

Now practice at your level:

fangmath.com $\rightarrow$ Geometry $\rightarrow$ Chapter 3